Zomato Restaurant Analytics Project

Project Documentation | SQL, PostgreSQL, pgAdmin 4 and Power BI

1. Introduction

This project analyzes restaurant-related data to explore restaurant distribution across cities, cuisine categories, food prices, and dining/delivery ratings. SQL is used for data exploration and analysis, while Power BI is used to present findings in an interactive dashboard.

2. Problem Statement

Restaurant datasets can contain many records and multiple attributes, making it difficult to compare cities, cuisines, prices, and ratings quickly. This project organizes the data-analysis workflow and presents useful summaries through SQL queries and visual reports.

3. Objectives

• Explore restaurant data and understand its structure.

• Use SQL queries to inspect and analyze the dataset.

• Compare restaurant counts across cities.

• Review cuisine distribution and food-price patterns.

• Compare dining and delivery ratings across cities.

• Identify top food items by price and restaurants by dining rating and votes.

• Build an interactive Power BI dashboard with City and Cuisine filters.

4. Tools and Technologies

• PostgreSQL — relational database for storing and querying data.

• pgAdmin 4 — interface for managing the PostgreSQL database and running SQL.

• SQL — data exploration and analytical queries.

• Microsoft Excel / CSV — dataset file format and initial data source.

• Microsoft Power BI Desktop — dashboard and visual reporting.

• GitHub — version control and project portfolio hosting.

5. Dataset Description

The project uses a Zomato restaurant dataset. The fields visible in the Power BI model include Best Seller, City, Cuisine, Delivery Rating, Delivery Votes, Dining Rating, Dining Votes, Item Name, Place Name, Prices, Restaurant Name, and Votes.

The repository may contain a raw dataset and a cleaned dataset. Keep the cleaned file only if it has genuinely been cleaned and validated. Preserve the original raw data separately.

Before publishing the dataset publicly, confirm that redistribution is permitted. If it is not permitted, upload the analysis files and document the dataset source without sharing the data file.

6. Data Preparation and SQL Analysis

SQL scripts are maintained in the SQL folder. They should document the actual queries used for data exploration, quality checks, cleaning, and analysis.

Recommended checks include inspecting column names and data types, checking missing values, checking duplicate records, validating numeric fields such as prices and ratings, and comparing row counts before and after cleaning.

Use the actual table and column names from the PostgreSQL database in the scripts. The SQL file should be tested against the project database before the repository is published.

7. Power BI Dashboard

The Power BI report includes KPI cards for Total Restaurants, Average Delivery Rating, and Average Food Price.

Visuals include Restaurant Count by City, Restaurant Distribution by Cuisine, Top 10 Expensive Food Items, Top 10 Restaurants by Dining Rating, Average Food Price by City, Dining vs Delivery Rating by City, a map visual, and Top 10 Restaurants by Votes.

City and Cuisine slicers allow users to filter the report. The dashboard screenshot and PDF export are included as static previews; the PBIX file is required to inspect the interactive report in Power BI Desktop.

8. Dashboard Metrics and Interpretation

The current report displays 825 for the Total Restaurants card, 3.96 for Average Delivery Rating, and 241.38 for Average Food Price. These values reflect the current Power BI model and its aggregation settings; verify the distinct-count logic and aggregation definitions before treating them as final business metrics.

The city chart shows the restaurant counts represented in the loaded dataset. The price chart ranks menu items by the sum of Prices, and the ratings chart ranks restaurants by average Dining Rating. The votes chart ranks restaurants by the Votes measure.

Use the slicers to compare results by city and cuisine. Interpret each chart according to its aggregation (count, average, or sum) and confirm that the field aggregation matches the intended business question.

9. Project Folder Structure

• Dataset/ — raw and, if validated, cleaned dataset files.

• SQL/ — SQL scripts used for data exploration, checks, cleaning, and analysis.

• PowerBI/ — Zomato_Restaurant_Analytics.pbix.

• Dashboard/ — dashboard_preview.png and Zomato_Dashboard.pdf.

• Documentation/ — this documentation file in DOCX and PDF formats.

10. How to Run the Project

1. Install PostgreSQL and pgAdmin 4 if they are not already installed.

2. Create or select the project database and review the SQL scripts.

3. Run the scripts in the correct order, adjusting table names or paths only where required by the actual schema.

4. Open the PBIX file in Microsoft Power BI Desktop.

5. If prompted, update the data-source path or connection settings to match your environment.

6. Use the City and Cuisine slicers to explore the report.

7. Review the PNG and PDF files for a quick static preview.

11. Limitations and Validation

Results depend on the source dataset, its quality, and the aggregation settings in Power BI.

The number displayed as Total Restaurants should be checked to ensure it represents distinct restaurants rather than item-level records.

The current chart titled Top 10 Expensive Food Items uses Sum of Prices; confirm that summing menu prices is the intended comparison, since an average price per item may answer a different question.

Map locations and geographic labels depend on the available city/place fields and Power BI map geocoding.

Only publish data and files that you are permitted to share. Never include database passwords, access tokens, or other credentials.

12. Conclusion

This project demonstrates an end-to-end restaurant analytics workflow using SQL and Power BI. It combines data exploration and analytical queries with an interactive dashboard to summarize city-level restaurant counts, cuisine distribution, prices, ratings, and votes. The project can be extended by adding validated data-quality results, documented business findings, and further analysis.

13. Repository Checklist

• Confirm that the SQL scripts run successfully.

• Verify that the raw and cleaned dataset files are correctly named and are actual files, not folders.

• Ensure the PBIX file opens and its data-source path is documented.

• Include the dashboard PNG and PDF preview.

• Review the README and this documentation for accuracy.

• Remove private credentials and confirm dataset-sharing permissions before making the repository public.